ALL INDIA TEST SERIES

FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

FULL TEST – II

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 30)** contains 30 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics PART – I

SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

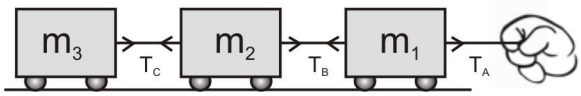
1. The physical quantity, electric potential (V) has dimensions $[M^a L^b T^c A^d]$ and capacitance (c) has dimensions $[M^e L^f T^g A^h]$. Then a + b + c + d + e + f + g + h is :
   (A) zero (B) 1
   (C) 2 (D) None of these

2. Mark the CORRECT statement:
   (A) Two vectors are equal to each other only if their corresponding components are equal
   (B) The cross product of two parallel vectors can be NON-zero
   (C) The direction of a zero vector can be determined
   (D) If we multiply $\vec{A}$ by the scalar-1, we obtain the vectors $-\vec{A}$, which is a vector with the opposite magnitude but the same direction as $\vec{A}$

3. The acceleration of a particle moving only on a horizontal x-y plane is given by $a = 2t\hat{i} + 3t\hat{j}$, where a is in meters per second squared and t is in seconds. At t = 0 the particle has the velocity vector $\vec{V} = (5.00\,m/s)\hat{i} + (2.00\,m/s)\hat{j}$ and position vector $\vec{r} = (10.00\,m)\hat{i} + (10.0m)\hat{j}$. At t = 6s, the position vector of the particle is :
   (A) $(112m)\hat{i} + (130m)\hat{j}$ (B) $(102m)\hat{i} + (120m)\hat{j}$
   (C) $(256m)\hat{i} + (238m)\hat{j}$ (D) None of these

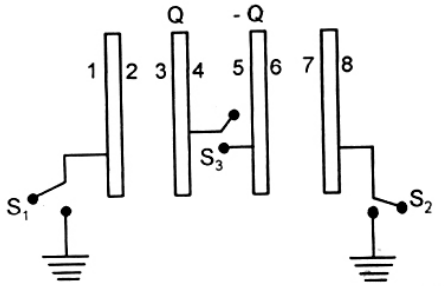
Space for Rough work

4. A child's toy consists of three cars that are pulled in tandem on small friction less rollers (Fig. A). The cars have masses m_1, m_2 and m_3 ($m_1 < m_2 < m_3$). If they are pulled to the right with a horizontal force P, mark the INCORRECT statement (T = Tension in the string):



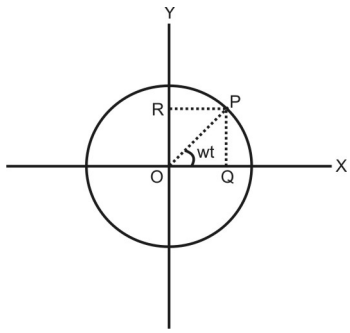
(A) $T_A > T_B > T_C$,
(B) $T_A < 2T_B$
(C) $T_B < 2T_C$
(D) $T_A > 3T_C$

5. Four conducting plates are arranged with equal spacing and surfaces of all the plates are marked as 1, 2, 3, 4, 5, 6, 7 and 8 as shown. Two middle plates are given charged +Q and –Q, after which S_1 and S_2 are closed. The charge distribution on the surfaces of conducting plates from 1 to 8 in order is:



(A) O, –Q, +Q, O, O, –Q, Q, O
(B) O, –Q/3, Q/3, 2Q/3, –2Q/3, –Q/3, O
(C) O, O, O, Q, –Q, O, O, O
(D) O, –2Q/3, 2Q/3, Q/3, –Q/3, –2Q/3, 2Q/3, O

6. Consider a particle P moving on a circle of radius 'A' with a constant angular speed 'w'. RP and PQ are $\perp$ to X and Y axes respectively:
(A) The x-coordinate of the particle at time t is A sin ωt
(B) The particle has a non-zero displacement in a time interval $2\pi/\omega$
(C) R and Q execute simple harmonic motion and their phases differ by $\pi/2$
(D) If the radius of the circle is doubled, the phase difference between R and Q would also double



Space for Rough work

7. An infinite non conducting uniformly charged sheet has a hole of radius 'R' in it (change density = σ). An electron is placed on an axis passing through the centre of hole and perpendicular to the plane of sheet at a distance 'R' from the centre. The speed with which the electron reaches the centre is ....

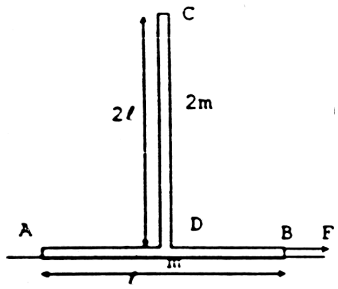
($\sigma > 0$) Given : electric field on the axis of a uniformly charged disc of radius 'r'

$$E(x) = \frac{\sigma}{2\in}\left[1 - \frac{x}{\sqrt{x^2 + r^2}}\right]$$

(A) $\sqrt{\frac{\sigma eR}{m\varepsilon_0}}$ (B) $\sqrt{\frac{2\sigma eR}{m\varepsilon_0}}$

(C) $\sqrt{\frac{\left(\sqrt{2}-1\right)\sigma eR}{m\varepsilon_0}}$ (D) $\sqrt{\frac{\sqrt{2}\sigma eR}{m\varepsilon_0}}$

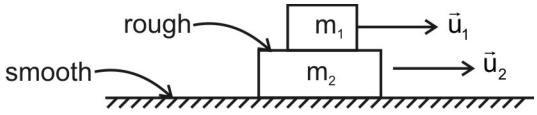
8. Two thin rigid rods, AB (mass m, length ℓ) and CD (mass 2m, length 2ℓ) are connected to make a 'T' as shown in figure. The assembly is kept vertical on a frictionless horizontal surface and is restrained to fall sidewise. What is the least value of F to be applied horizontally at B, such that B is able to leave contact with the surface :



(A) 3mg (B) (9/4) mg

(C) $\frac{3\sqrt{3}}{2}$mg (D) $\frac{3}{2}$mg

9. In a two block system shown in the figure, the two blocks have velocities in same horizontal direction. Mark the CORRECT statement about 'the velocity of the centre of mass' of the system with respect to a stationary frame:



(A) $\vec{V}_{cm}$ is a function of time and will become $\vec{O}$ at some instant(s)

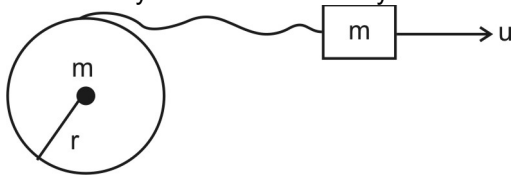
(B) $\vec{V}_{cm}$ is non-zero at all times

(C) $\vec{V}_{cm}$ is $\vec{O}$ at all times

(D) $\vec{V}_{cm}$ may be $\vec{O}$ at some instant(s)

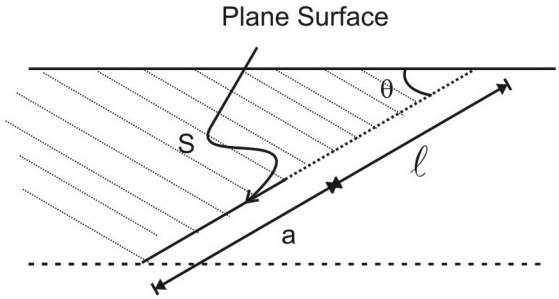
Space for Rough work

10. A block of mass m is attached to a pulley disc of same mass and radius 'r' by means of a block strings a shown. The pulley is hinged about its centre on a horizontal table and the block is projected with an initial velocity of 3 m/s. Its velocity when the string becomes taut will be:



(A) 3 m/s
(B) 2 m/s
(C) 1 m/s
(D) 2.5 m/s

11. Consider a plane surface of length 'a' and width 'b' which is inside a liquid of density ρ. Find the force on the entire surface due to the liquid in S.

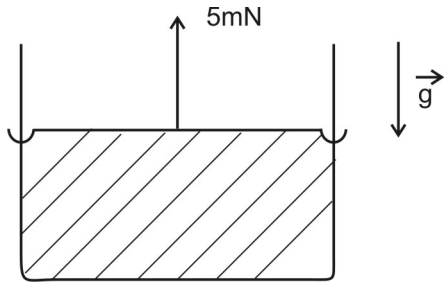


(A) $\frac{\rho gab}{2}[a+2\ell]\sin\theta$
(B) $\frac{\rho gab}{2}[a+\ell]\sin^2\theta$
(C) $\frac{\rho gab}{2}[a+\ell]\sin^2\theta$
(D) $\frac{\rho gab}{4}[a+2\ell]\sin^2\theta$

12. A pressure meter attached to a closed water tap reads 1 bar. When the tap speed is 10 ms^{-1} and the reading of the pressure meter is
(A) 1.5×10^5 Pa
(B) 3×10^5Pa
(C) 0.5×10^5 Pa
(D) 10^5 Pa

Space for Rough work

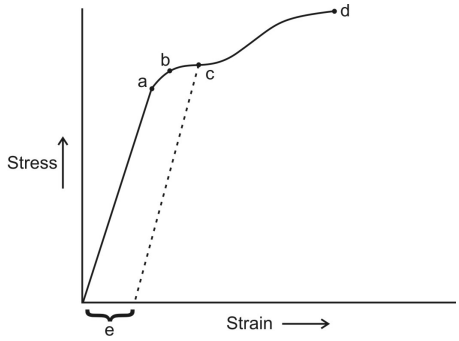
13. A film of the liquid is supported in a vertical rectangular area, the top border of which is a sliding wire. An external force of 5 mN acts on the sliding wire of length 50 mm and linear mass density of 1.75×10^{-3} kg/m. Find the surface tension of the liquid:



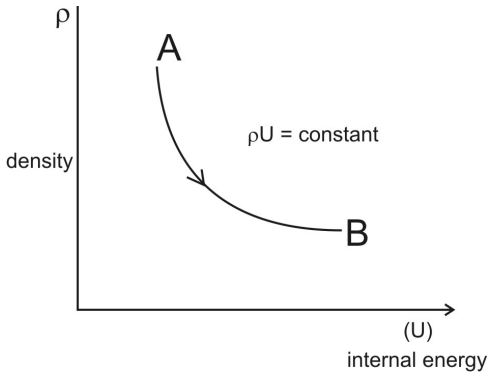
(A) 0.083 N/m
(B) 0.041 N/m
(C) 0.1 N/m
(D) 0.05 N/m

14. The magnitude of gravitational potential energy of the sun-earth system is u with zero potential energy at infinity separation. The kinetic energy of the earth with respect to the sum is K.
(A) U < K
(B) U > K
(C) U = K
(D) None of these

15. Mark the CORRECT option, for stress-strain curve of a metal wire:
(A) Curve ab represents elastic behaviour and e denotes fracture
(B) ONLY curve ac represents elastic behaviour
(C) Curve bd represents plastic behaviour and e represents permanent set
(D) Point d represents fracture point and the gap between b and d is respectively large for brittle materials



16. Mark the CORRECT option for a gas:
(A) Heat is expelled by gas
(B) Heat is absorbed by gas
(C) Process A→B is adiabatic
(D) Temperature of the gas decreases from A to B



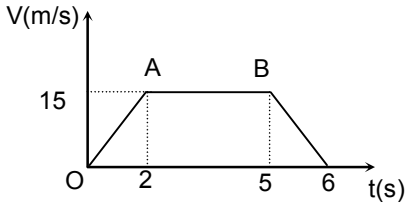
Space for Rough work

17. A musical instrument has a string which is timed to frequency 400 Hz and has a length of 0.5m. What is the third longest wavelength of resonance that reaches the ear of the listener? (V_{air} = 343 m/s):
(A) 0.572 m (B) 0.286 m
(C) 1.7515 m (D) 1.29 m

18. The most probable speed of a gas with temperature 'T' and molar Mass 'M' is $\sqrt{\frac{ART}{M}}$, A is a dimensionless constant and R is universal gas constant in $JK^{-1}\ mol^{-1}$. Four equal charges $A\times10^{-6}$C each are fixed at the four corners of a square of side 5 cm. Find the coulomb force experienced by one of the charges due to the other three :
(A) 20 N (B) 27.5 N
(C) 44.6 N (D) 61.9 N

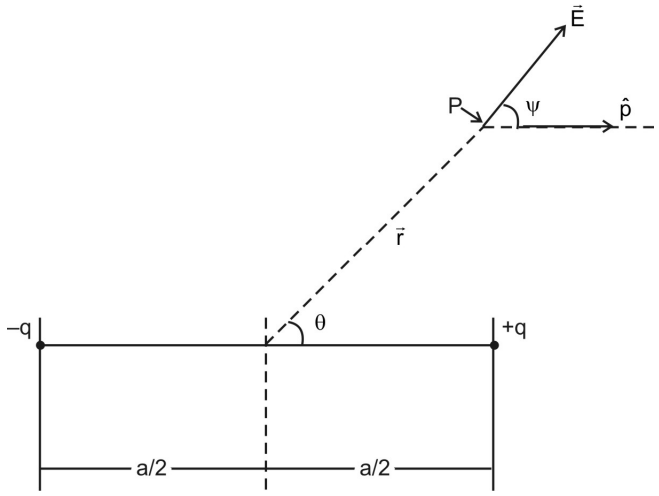
19. One cannot see through fog, because:
(A) Fog absorbs the light
(B) Light suffers total reflection at droplets
(C) Refractive index of the fog is infinity
(D) Light is scattered by droplets

20. The speed time graph of a car is given below the car weights 1 Mg.
(A) Distance travelled in first two seconds is 30 m
(B) Acceleration of the car at t = 4s is 7.5 m/s^2
(C) Braking force applied at the end of 5s to bring the Car to a stop within 1s is 15 KN
(D) (A), (B) and (C) are WRONG



Space for Rough work

21. Electric field due to a 'short' electric dipole at a point P as shown is at an angle ψ with respect to $\vec{p}$. Then



(A) $\psi = \theta + \tan^{-1}\left(\frac{\tan\theta}{\sqrt{2}}\right)$ (B) $\psi = \theta + \tan^{-1}\left(\frac{\tan\theta}{2}\right)$

(C) $\psi = \theta + \tan^{-1}\left(\frac{\tan\theta}{2\sqrt{2}}\right)$ (D) $\psi = \theta + \tan^{-1}\left(\frac{\tan\theta}{\sqrt{5}}\right)$

22. Mark the INCORRECT statement:
(A) By convention, the charge on a plastic rod rubbed with fur is negative and the charge on a glass rod rubbed with silk is positive.
(B) The ratio of electric force and gravitational force between a proton and electron is $\frac{ke^2}{Gmemp} \cong 2.4 \times 10^{39}$
(C) Conservation of total charge of an isolated system is a property dependent on the scalar nature of charge.
(D) Quantisation of electric charge is a basic unexplained law of nature, interestingly; there is no analogues law of quantisation of mass.

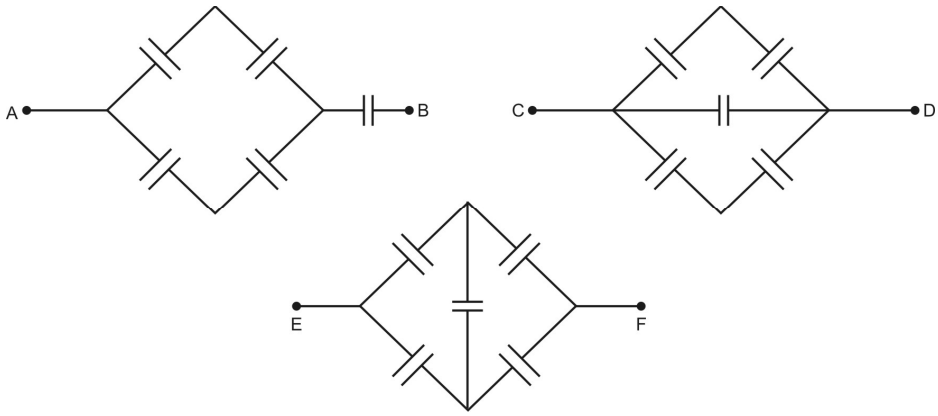
Space for Rough work

23. A substance when placed in a non-uniform magnetic field is attracted towards stronger magnetic field. That substance CAN be:
(A) Paramagnetic
(B) Diamagnetic
(C) Either Paramagnetic OR diamagnetic
(D) Neither Paramagnetic NOR diamagnetic

24. Mark the CORRECT option:
(A) A vertical plane containing the vertical axis (orthogonal to ground) and the longitude is called the geographic meridian
(B) Geographic meridian and magnetic meridian for earth are always same at a point
(C) The dip circle is simply a compass needle pivoted about a fixed vertical axis and free to move in a horizontal plane
(D) There is a bar magnet busied deep inside earth's interior which gives rise to earth's magnetic field

25. Two resistors R and 2R are connected in parallel in an electric circuit. The thermal energy developed in R and 2R are in the ratio:
(A) 1 : 2 (B) 2 : 1
(C) 1 : 4 (D) 4 : 1

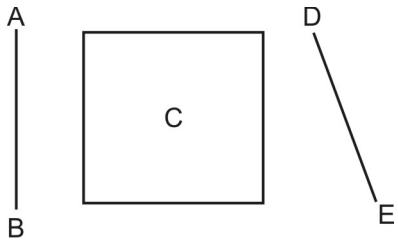
26. Consider the circuits drawn below:



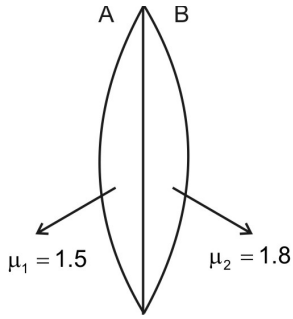
All the capacitors are identical, then
(A) $C_{CD} > C_{EF} = C_{AB}$ (B) $C_{CD} > C_{EF} > C_{AB}$
(C) $C_{CD} > C_{AB} > C_{EF}$ (D) $C_{EF} > C_{CD} > C_{AB}$

Space for Rough work

27. A wave front AB passing through a system C averages as DE. The system 'C' CAN be
(A) a slit
(B) a biprism
(C) a prism
(D) a glass slab



28. A convex lens is made of two different materials having radius of curvature of surface A and surface B as 25 cm and 24 cm respectively. The effective focal length of lens is
(A) 75/4 cm
(B) 75 cm
(C) 50 cm
(D) 30 cm



29. Hydrogen atom is its ground state is excited by means of monochromatic radiation of wavelength 970.6Å. How many different wavelengths are possible in the resulting EMISSION spectrum? [hc = 1.986×10^{-25} J-m] :
(A) 3 (B) 4
(C) 5 (D) 6

30. Attenuation, when referring to communication systems, is :
(A) The process of mixing low frequency signal and a high frequency signal
(B) The weakening of a signal as it propagates through a medium
(C) The process of increasing the amplitude of a signal
(D) The process of recovering a low frequency signal from a high frequency signal

Space for Rough work

Chemistry

PART – II

SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. A particular 100-octane aviation gasoline used 1.00 cm^3 of tetraethyl lead $(C_2H_5)_4Pb$ of density 1.66 g/cc, per litre of product. This compound is made as follows.
$4C_2H_5Cl + 4NaPb \longrightarrow (C_2H_5)_4Pb + 4NaCl + 3Pb$. How many gram of ethyl chloride is needed to make enough tetraethyl lead for 1 L of gasoline.
(A) 1.33g (B) 2.66 g
(C) 9.2g (D) 0.33g

2. The treatment of $CH_3CH = CHCH_3$ with $NaIO_4$ produces
(A) CH_3CHO only (B) CH_3COOH only
(C) CH_3CHO & CH_3COOH (D) CH_3COCH_3 + HCOOH

3. Among the following, which exhibits aromaticity?
(A) Cyclopentadienyl cation (B) Cyclopentadienyl radical
(C) Cycloheptatrienyl radical (D) Cycloheptatrienyl cation

4. S_N1 characteristics would be exhibited in the hydrolysis of
(A) chlorobenzene (B) 1-chloronapthalene
(C) propenyl chloride (D) 2-chloro-1-propene

5. The ease of dehydration of the following alcohols with H_2SO_4 is
(I) $CH_3CH_2CH(OH)CH_2CH_2CH_3$ (II) $(CH_3)_2C(OH)CH_2CH_2CH_3$
(III) $CH_3CH_2CH(OH)CH(CH_3)_2$ (IV) $(CH_3)_2C(OH)CH(CH_3)_2$

(A) III > IV > II > I (B) III > II > IV > I
(C) II > III > I > IV (D) II > III > IV > I

Space for Rough work

6. The solubility product of A_2X_3 is 1.08×10^{-23}. Its solubility will be
(A) 1×10^{-3} (B) 1×10^{-6}
(C) 1×10^{-4} (D) 1×10^{-5}

7. When dilute H_2SO_4 is electrolysed between Pt electrodes, the gas liberated at the anode will be
(A) SO_2 (B) SO_3
(C) O_2 (D) H_2

8. 0.7 g of $Na_2CO_3 .xH_2O$ were dissolved in water and the volume was made to 100 mL. 20 mL of this solution required 19.8 mL of N/10 HCl for complete neutralization. The value of x is
(A) 7 (B) 2
(C) 3 (D) 5

9. The product of reaction of an aq. solution of Bi^{3+} salt with sodium thiosulphate gives
(A) Bi_2S_3 (B) $Na_3[Bi(S_2O_3)_3]$
(C) $Na[Bi(S_2O_3)_2]$ (D) $[Bi_2(S_2O_3)_2]Cl_2$

10. The basic ionisation constant for hydrazine, N_2H_4 is 9.6×10^{-7}. What would be the percent hydrolysis of 0.1 N_2H_5Cl?
(A) 0.016% (B) 3.2%
(C) 1.6% (D) 0.032%

Space for Rough work

11. Which of the following combinations will act as a buffer
(A) NH_3 + HCl (molar ratio 2:1) (B) HC_2H_3O, NaOH (Molar ratio 1:2)
(C) HCl + NaCl (D) NaOH + $HC_2H_3O_2$ (1:1 molar ratio)

12. Gaseous cyclobutane isomerizes to butadiene in a first order process which has a k value at 153°C of $3.3 \times 10^{-4}s^{-1}$. How many minutes would it take for the isomerisation to proceed 40% to completion at this temperature?
(A) 26 min (B) 52 min
(C) 13 min (D) None of these

13. Current efficiency is defined as the extent of a desired electrochemical reaction divided by the theoretical extent of the reaction times 100%. What is the current efficiency of an electrodeposition of Cu metal in which 9.8 g Cu is deposited by passage of 3A current for 10000 seconds.
(A) 0.9% (B) 49.9%
(C) 99.1% (D) 51.1%

14. A 10 L cylinder of oxygen at 4 atm pressure and 17°C developed a leak. When the leak was repaired, 2.5 atm of O_2 remained in the cylinder, still at 17°C. The no. of moles of gas escaped.
(A) 1.2 mol (B) 0.30 mol
(C) 0.63 mol (D) 2.4 mol

15. Arrange the following in decreasing order of basicity
(I) Aniline (II) p-nitro aniline (III) N, N dimethyl aniline (IV) N, N, 2,6 tetramethylaniline

(A) I >II > IV > III (B) III > IV > I > II
(C) III > I > IV > II (D) IV > III > I > II

Space for Rough work

16. Which of the following is not linear
(A) $Ag(NH_3)_2^+$ (B) $HgCl_2$
(C) $PbCl_2$ (D) CS_2

17. The order of enol content in the following compounds is

CH_3COCH_2COOEt	$(EtOCO)_2CH_2$	$PhCOCH_2COCH_3$	$PhCOCH_2COOEt$
(I)	(II)	(III)	(IV)

(A) II > I > III > IV (B) IV > III > II >I
(C) III > IV > I > II (D) III > I > IV > II

18. All the halobenzenes are separately treated with HNO_3/H_2SO_4. The quantity of o-product should be maximum when the substance is
(A) Flurobenzene (B) Chlorobenzene
(C) Bromobenzene (D) Iodobenzene

19. The minimum number of carbon atoms for an alkene hydrocarbon to exhibit optical isomerism is
(A) 5 (B) 6
(C) 9 (D) None of these

20. Bicyclohexane was found to undergo two parallel 1st order rearrangements. At 730 K the first order rate constant for the formation of cyclohexene was 1.26×10^{-4} s^{-1} and for the formation of methyl cyclopentene the rate constant was 3.8×10^{-5} s^{-1}. What was the percentage distribution of methyl cyclopentene.
(A) 77% (B) 23%
(C) 66% (D) 34%

21. $\Delta T_f / K_f$ has the same value of 1 mol kg^{-1} for 8% AB_2 and 10% A_2B by mass of solvent, both AB_2 and A_2B being non electrolytes.
Atomic masses of A and B will be respectively.
(A) 20, 40 (B) 20, 50
(C) 40, 20 (D) 50, 40

22. What will happen when D-(+)-glucose is treated with methanolic HCl followed by Tollen's reagent?
(A) A black ppt. will be formed. (B) A green colour will appear.
(C) A red ppt. will be formed. (D) No characteristic colour or ppt. will be formed.

Space for Rough work

23. The r^+ / r^- ratio of ZnS is 0.402. Which of the following is the incorrect statement?
(A) ZnS is 4 : 4 coordination compound.
(B) ZnS does not crystallize in rock salt type lattice because r^+ / r^- is too small to avoid overlapping of S^{2-}.
(C) Zn^{2+} ion is too small to fit precisely into the tetrahedral voids of S^{2-} ions.
(D) Zn^{2+} ion is too large to fit into tetrahedral voids of S^{2-} ions.

24. The wavenumber of last line of Lyman series of hydrogen spectrum is 109674 cm^{-1}. The wave number of $H\alpha$ line in Balmer series He^+ is?
(A) 438696 cm^{-1} (B) 106974 cm^{-1}
(C) 30465 cm^{-1} (D) 60930 cm^{-1}

25. Consider the following statements
(I) NO^+ is more stable towards dissociation into atoms than NO.
(II) CO^+ is less stable towards dissociation into atoms than CO.
Then which of the following is correct?
(A) I only (B) II only
(C) I & II both (D) None of these

26. Below critical micelle concentration (CMC) sodium oleate in aqueous solution:
(A) exists largely as micelles of anions.
(B) dissolve substances like grease, fat etc. colloidally.
(C) increases the viscocity of water abruptly.
(D) behaves as strong electrolyte.

Space for Rough work

27. Given that:
$S^{o}_{H_2} = 131\ JK^{-1}\ mol^{-1}$
$S^{o}_{Cl_2} = 223\ JK^{-1}\ mol^{-1}$
and $S^{o}_{HCl} = 187\ JK^{-1}\ mol^{-1}$
The standard entropy change in formation of 1 mole of HCl (g) from H_2(g) and Cl_2(g) will be:
(A) 20 J K^{-1} (B) 10 J K^{-1}
(C) 187 J K^{-1} (D) 374 J K^{-1}

28. The hybridization of P in PO_4^{3-} is the same as that of:
(A) N in NO_3^- (B) S in SO_3
(C) I in ICl_2^+ (D) I in ICl_4^-

29. Among the following compounds which exist as solid at room temperature?
(A) XeF_6 (B) OF_2
(C) SF_4 (D) SF_6

30. Magnesium is mainly extracted by
(A) Carbon reduction method
(B) self reduction method
(C) the method of electrolysis
(D) leaching with aq. solution of NaCN followed by reduction

Space for Rough work

Mathematics

PART – III

SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. If the lines represented by $x^2 - 2pxy - y^2 = 0$ are rotated about origin through an angle θ, one in clockwise direction and other in anticlockwise direction, then equation of the bisectors of the angle between the lines in the new position is
(A) $px^2 + 2xy + py^2 = 0$ (B) $px^2 - 2xy + py^2 = 0$
(C) $px^2 - 2pxy - py^2 = 0$ (D) $px^2 + 2xy - py^2 = 0$

2. Let $D_0 = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ and let D_1 denote the determinant formed by the co-factor of elements of D_0 and D_2 denote the determinant formed by co-factor of D_1 and so on. If D_n denotes the determinant formed by the co-factor of D_{n-1}, then the value of determinant D_n is
(A) $(D_0)^{2n}$ (B) $D_0^{2^n}$
(C) $D_0^{n^2}$ (D) D_0^2

3. The maximum value of $f(x) = \int_0^1 t\sin(x + \pi t)dt$ is
(A) $\frac{1}{\pi}\sqrt{\pi^2 + 4}$ (B) $\frac{1}{\pi^2}\sqrt{\pi^2 + 4}$
(C) $\sqrt{\pi^2 + 4}$ (D) $\frac{1}{2\pi^2}\sqrt{\pi^2 + 4}$

Space for rough work

4. Number of integers in the range of 'a' so that the equation $x^3 - 3x + a = 0$ has all its roots real and distinct, is
(A) 2 (B) 3
(C) 4 (D) 5

5. $f(n) = \cot^2\frac{\pi}{n} + \cot^2\frac{2\pi}{n} + ... + \cot^2\frac{(n-1)\pi}{n}$, $(n > 1, n \in N)$ then $\lim\limits_{n\to\infty}\frac{f(n)}{n^2}$ is equal to
(A) $\frac{1}{2}$ (B) $\frac{1}{3}$
(C) $\frac{2}{3}$ (D) 1

6. Given two circles $x^2 + y^2 + 3\sqrt{2}(x+y) = 0$ and $x^2 + y^2 + 5\sqrt{2}(x+y) = 0$, let the radius of a circle which touches the two circles and their common diameter be $\frac{2\lambda - 1}{\lambda}$ then λ equals
(A) 10 (B) 8
(C) 7 (D) 5

7. Let ABCD be a tetrahedron with AB = 41, BC = 36, CA = 7, DA = 18, DB = 27 and DC = 13, then square of the distance between mid-points of AB and CD equals
(A) 1096 (B) 548
(C) 274 (D) 137

8. In triangle ABC, if 2b = a + c and A – C = 90º, then sin B equals
(A) $\frac{\sqrt{7}}{5}$ (B) $\frac{\sqrt{5}}{8}$
(C) $\frac{\sqrt{7}}{4}$ (D) $\frac{\sqrt{5}}{3}$

9. The graph of $f(x) = x^2$ and $g(x) = cx^3$ intersect at two points. If area of region bounded between f(x) and g(x) over the interval $\left[0, \frac{1}{c}\right]$ is equal to $\frac{2}{3}$ then the value of $\left(\frac{1}{c} + \frac{1}{c^2}\right)$ is
(A) 20 (B) 2
(C) 6 (D) 12

Space for rough work

10. The value of sec 40º + sec 80º + sec 160º will be
(A) 4 (B) –4
(C) 6 (D) 8

11. Consider the chords of the parabola $y^2 = 4x$ which touches the hyperbola $x^2 – y^2 = 1$, the locus of the point of intersection of the tangents drawn to the parabola at the extremities of such chords is a conic section having latus rectum λ, the value of λ is
(A) 1 (B) 2
(C) 3 (D) 4

12. The area of the region bounded by lines y = x, y = 0 and $x = \sin^{-1}(a^4 + 1) + \cos^{-1}(a^4 + 1) – \tan^{-1}(a^4 + 1)$ is
(A) $\frac{\pi^2}{8} - \frac{a^2}{4}$ (B) $\frac{\pi^2}{8} - \frac{a^2}{2}$
(C) $\frac{\pi^2}{16}$ (D) $\frac{\pi^2}{32}$

13. If $ax^3 + by^3 + cx^2y + dxy^2 = 0$, represents three distinct straight lines, such that each line bisects the angle between the other two, then which of the following is true
(A) 3b + c = 0 (B) 3a + d = 0
(C) d + 3a = 0 (D) b + 3c = 0

14. The number of integral value of n (where $n \geq 2$) such that the equation $2n\{x\} = 3x + 2[x]$ has exactly 5 solution (where [.] denotes the greatest integer function and {x} is fractional part of x) is
(A) 2 (B) 3
(C) 4 (D) 0

15. The rth, s^{th}, t^{th} terms of an A.P. are 6, 8 and 12 respectively. If $f(x) = tx^2 + 2rx – 2s$ then f(x) = 0 has
(A) both roots are negative (B) both roots greater then 2
(C) one root negative other greater than 1 (D) exactly one root $\in$ (0, 1)

16. Let a, b, c, d $\in$ R be such that $a^2 + b^2 + c^2 + d^2 = 25$, then
(A) $ab + bc + cd + ad \leq \frac{25}{2}$ (B) $ab + bc + cd + ad \leq 25$
(C) $ab + bc + cd + ad \leq 5$ (D) $ab + bc + cd + ad \leq \frac{5}{2}$

Space for rough work

17. Let $f(x)=\begin{cases}3\sin x+a^2-10a+30 & x\notin Q\\ 4\cos x & x\in Q\end{cases}$ which one of the following statements is correct?

(A) f(x) is continuous for all x when a = 5

(B) f(x) must be discontinuous for all x, when a = 5

(C) f(x) is continuous for all $x = 2n\pi - \tan^{-1}\left(\frac{3}{4}\right)$, $n \in I$, when a = 5

(D) f(x) is continuous for all $x = 2n\pi - \tan^{-1}\left(\frac{4}{3}\right)$, $n \in I$, when a = 5

18. Number of ways in which 5 alike red balls and 6 alike green balls can be arranged so that exactly two pairs of green balls are together is

(A) 72 (B) 90

(C) 144 (D) 100

19. If $|z-2+i| = |z|\left|\sin\left(\frac{\pi}{4}-\arg z\right)\right|$, then locus of z is

(A) pair of straight line (B) circle

(C) parabola (D) ellipse

20. If x, y, z are real variables such that 3 tan x + 4 tan y + 5 tan z = 20, then the minimum value of $\tan^2 x + \tan^2 y + \tan^2 z$ is

(A) 10 (B) 15

(C) 8 (D) 12

21. If $\theta = \frac{2\pi}{2009}$ then $\cos\theta\cos 2\theta \cos 3\theta$ ….. $\cos 1004\theta$ is equal to

(A) 0 (B) $\frac{1}{2^{2008}}$

(C) $\frac{1}{2^{1004}}$ (D) $-\frac{1}{2^{1004}}$

Space for rough work

22. $\int \frac{xdx}{\sqrt{1+x^2+\sqrt{\left(1+x^2\right)^3}}}$ is equal to

(A) $\frac{1}{2}\ell n\left(1+\sqrt{1+x^2}\right)+c$

(B) $\frac{-2}{3\left(1+\sqrt{1+x^2}\right)^{3/2}}+c$

(C) $2\left(1+\sqrt{1+x^2}\right)+c$

(D) $2\sqrt{1+\sqrt{1+x^2}}+c$

23. There exist two points P and Q on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ such that OP $\perp$ OQ (where O is origin), then number of points in the xy plane from where pair of perpendicular tangents can be drawn to the hyperbola is

(A) 0
(B) 1
(C) 2
(D) infinite

24. The shortest distance between the lines 2x + y + z – 1 = 0 = 3x + y + 2z – 2 and x = y = z is

(A) $\frac{1}{\sqrt{2}}$

(B) $\sqrt{2}$

(C) $\frac{3}{\sqrt{2}}$

(D) $\frac{\sqrt{3}}{2}$

25. If $A=\alpha\begin{bmatrix}1 & 1+i\\ 1-i & -1\end{bmatrix}$ ($\alpha \in R$) is a unitary matrix then α^2 is

(A) $\frac{1}{2}$

(B) $\frac{1}{3}$

(C) $\frac{1}{4}$

(D) $\frac{2}{9}$

Space for rough work

26. Solution of equation (2 + x) dy – y dx = 0 represents a curve passing through a fixed point p, then are of equilateral triangle with p as one vertex and x + y = 0 as its one side is

(A) $2\sqrt{3}$ (B) $\sqrt{3}$

(C) $\frac{2}{\sqrt{3}}$ (D) $\frac{4}{\sqrt{3}}$

27. The term independent of x in the expansion of $\left(\frac{1}{x^2}+\frac{1}{x}+1+x+x^2\right)^5$ is

(A) 381 (B) 441

(C) 439 (D) 359

28. If the standard deviation of the data 1, 3, 5, 7, ….., 2013 is d, then [d] is (where [.] denotes the greatest integer function)

(A) 580 (B) 581

(C) 582 (D) 583

29. If number of different reflexive relation on set A is equal to number of different symmetric relation on set A, then n(A) is equal to

(A) 1 (B) 3

(C) 1 and 3 (D) 3 and 7

30. If $(\sin^{-1} x + \sin^{-1} y)(\sin^{-1} z + \sin^{-1} w) = \pi^2$, then the value of $\begin{vmatrix} x^{n_1} & y^{n_2} \\ z^{n_3} & w^{n_2} \end{vmatrix}$ can not be equal to (where $n_1, n_2, n_3, n_4 \in N$)

(A) –2 (B) 0

(C) 1 (D) 2

Space for rough work